

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$f(x) = (x + 4)(x - 1)(2x - 3)$

The function f is defined above. Which of the following is NOT an x -intercept of the graph of the function in the xy -plane?

Answer

A. $(-4, 0)$

B. $(-\frac{2}{3}, 0)$

C. $(1, 0)$

D. $(\frac{3}{2}, 0)$

Correct Answer: B

Rationale

Choice B is correct. The graph of the function f in the xy -plane has x -intercepts at the points (x, y) , where $y = f(x) = 0$. Substituting 0 for $f(x)$ in the given equation yields $0 = (x + 4)(x - 1)(2x - 3)$. By the zero product property, if $0 = (x + 4)(x - 1)(2x - 3)$, then $x + 4 = 0$, $x - 1 = 0$, or $2x - 3 = 0$. Solving each of these linear equations for x , it follows that $x = -4$, $x = 1$, and $x = \frac{3}{2}$, respectively. This means that the graph of the function f in the xy -plane has three x -intercepts: $(-4, 0)$, $(1, 0)$, and $(\frac{3}{2}, 0)$. Therefore, $(-\frac{2}{3}, 0)$ isn't an x -intercept of the graph of the function f .

Alternate approach: Substitution may be used. Since by definition an x -intercept of any graph is a point in the form $(k, 0)$ where k is a constant, and since all points in the options are in this form, it need only be checked whether the points in the options lie on the graph of the function f . Substituting $-\frac{2}{3}$ for x and 0 for $f(x)$ in the given equation yields $0 = (-\frac{2}{3} + 4)(-\frac{2}{3} - 1)(2(-\frac{2}{3}) - 3)$, or $0 = \frac{650}{27}$. Therefore, the point $(-\frac{2}{3}, 0)$ doesn't lie on the graph of the function f and can't be an x -intercept of the graph.

Choices A, C, and D are incorrect because each of these points is an x -intercept of the graph of the function f in the xy -plane. By definition, an x -intercept is a point on the graph of the form $(k, 0)$, where k is a constant. Substituting -4 for x and 0 for $f(x)$ in the given equation yields $0 = (-4 + 4)(-4 - 1)(2(-4) - 3)$, or $0 = 0$. Since this is a true statement, the point $(-4, 0)$ lies on the graph of the function f and is an x -intercept of the graph. Performing similar substitution using the points $(1, 0)$ and $(\frac{3}{2}, 0)$ also yields the true statement $0 = 0$, illustrating that these points also lie on the graph of the function f and are x -intercepts of the graph.

